

Artificial Intelligence and Automation and its Effect on Foreign Labor

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Abstract: This paper studies artificial intelligence's (AI) impact on skilled and unskilled immigrant labor in the United States (U.S.). By utilizing other studies, papers, graphs, and data the impact of AI on foreign born workers can be illustrated. Based on the findings and trends, we discussed policy reformations and laws. These policy ideas will help lessen the friction between AI and foreign-born workers.

Keywords: Artificial intelligence, Foreign Workers, Immigrant, Labor

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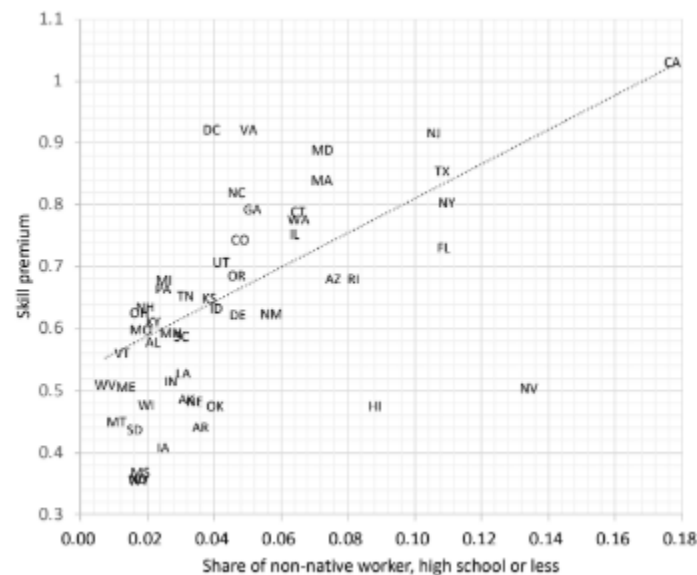
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Introduction

Artificial intelligence (AI) and automation are at the forefront of today's society. With rapid growth in many industries, there have been many benefits when it comes to product optimization. However, in contrast to the many benefits of its application, many people suffer due to these technologies. These people suffering would be those whose skill sets do not complement the new technological age or those who have just replaced and faded out with the implementation of these technologies. The main examples are the new developments in Amazon-backed companies and Tesla's new Optimus robot. As recently as April 2024, Agility Robotics, which is an Amazon-backed robotics company, laid off an unspecified but claimed it was a small number of employees. Also, on June 12, Tesla deployed two robots onto their factory floor. The new age is upon us, and here are some things we can do to prevent a total replacement of the foreign-born labor force that makes up many of these factories.

While AI does benefit people with specialized skill sets, there is an expectation that uneducated and low-skill workers might be replaced in the labor force, causing a decrease in the skill premium and possibly immigration. This assumption is available because seen in other research papers, graphs, and analytical studies have been conducted to show the positive correlation between low-skill immigration and the skill premium; hence why, if this premium is to decrease, a decrease in low-skill immigrant labor is expected (Mandelman et., 2024). As shown in Figure 1 from Mandelman et al. (2024), there are correlations between skill premium and immigration. As immigration increases or foreign labor, when immigrants with only education up to high school or less are in the workforce, the skill premium increases in most states.

Figure 1: Skill premium and its positive correlation with low-skill immigration in different states



Source: Federico S. Mandelman, Yang Yu, Francesco Zanetti, and Andrei Zlate. (January 2024) “Slowdown in Immigration, Labor Shortages, and Declining Skill Premia”, Figure 3, Federal, Reserve Bank of Atlanta, Working Paper Series.

There are many problems that can stem from this. The decrease in immigration lowers the workforce in the U.S. and impacts many areas, such as skill premium, decreased stock of immigration, and a decrease in low-skill employment (Mandelman et al., 2024). Most importantly, this reduction in skill premium can reduce incentives to train native workers, which will cause a decrease in productivity. A decrease in productivity will, in tandem, cause a decrease in welfare, decreasing the quality of life for not just foreign workers but also the citizens of the U.S., causing possible economic frictions.

These job frictions have been considered in some academic papers. As seen in “*Can massive technological progress hurt workers? A review of Power and Progress*” by Daron Acemoglu and Simon Johnson, criticism has been that so-called ordinary people, on average, will not be able to compete in this new technological age and will be replaced by machines. This

competition is mostly due to lacking skills or interest in the subject. Most authors have stated that the way to combat these frictions is to implement policies. The policies in this case included breaking big tech and more regulation, improving competition, and bettering the job market. However, Daron and Simon are not the only researchers who have analyzed these job frictions.

This article contributes to existing literature in two ways. First, it presents and summarizes empirical facts about AI and the labor force. We achieve that goal by using data from the Saint Louis Federal Reserve (FRED), the Census Bureau (CB), the Migration Policy Institute (MPI), and the Bureau of Labor Statistics (BLS). We show how the number of immigrants in the US has been increasing since 1850 but at a lower rate since 2000, and the labor force participation of immigrants has declined in the last couple of years. Second, we provide policy recommendations to address some of the issues found in the paper. We conclude with these statements and possible future policy and economic research to test these provided recommendations.

This paper is organized as follows: Section 2 summarizes the literature. Section 3 describes the data. Section 4 shows our main results. Lastly, Section 5 concludes.

Literature Review

With the rapid growth of Artificial intelligence (AI) and automation, several papers have studied their possible impacts on labor, immigration, and welfare. Furthermore, public debt, technological innovations creating externalities, and uninsured markets (displaced workers) leave space for policy recommendations. In this section, we will summarize the crucial findings that relate to our work.

Mandelman and Zlate (2020) show that since the early 1980s, many jobs have been created for the U.S. people; however, despite this, at the lower tail of the skill distribution charts, there is

a high amount of immigration. By implementing several data series that model innovations in offshoring, automation, and low-skill immigration in the last 30 years, they find that offshoring has caused the market to become polarized in tandem with recent improvements to automation. They speculate that factors will shape the future of polarization in the workforce. This is due to the recent spike in high-skill immigration, changing the spread of labor in the U.S. for the foreseeable future and possibly causing frictions.

Beraja and Zorzi (2023) study automation, and they analyze friction between workers and the recent implementation of mass automation at many firms. They find that many firms fail to internalize said automation, displacing some workers and lowering income for others when they become constrained from borrowing. They have also found that the government sees the slowing down of optimization as beneficial because of efficiency despite the lack of equity in this practice. Suggesting two possible directions that should be taken in the future. First, an optimal policy should be found to fix the friction between workers and automation. Secondly, another possible direction is to create a quantitative model that focuses on answering other policy questions dealing with workers' displacement in different industries.

Bloom et al. (2023) analyze the new growing age and the changing skill premium as services such as ChatGPT continue to grow. They conclude that automation and “robots” will continue to substitute low-skill workers while, as AI develops, there will be a higher increase in the number of high-skill workers replaced. This means that AI has a heavy potential to decrease the skill premium, reversing the increases in inequality, so while not all bad, it could decrease productivity, as stated before. They then conclude that it would be beneficial to study the elasticities of substitution of low and high-skill workers with AI in the future. This would then allow for a better understanding of the positives and negatives of this new technological age.

Colombo et al. (2019) have also investigated what it has been like since AI has met the labor market in their research. They find that there are many benefits to AI and that they will create tools to shape the future of analytics in all fields. However, they can see through tables that many jobs may be at risk of becoming automated, replacing people in the labor force. There are many complements to using AI, and much of the negative impact of this technology is in the manufacturing fields. Shapiro and Mandelman (2019) go over the new digital age as well as newly implemented automation in the labor force. They notice that in these environments there are extremely large gaps in role barriers at salaried firms. They also provide better understanding by representing the relationship that technology and automation hold in the current market and how it facilitates labor competition.

Chodavadia et al. (2024) review immigrant entrepreneurship correlation with innovation. They find that there is indeed a positive relationship and suggest that immigrant workers are 3.4-4.5 percent more likely to establish new technology. Along with this other research from Mandelman et al. (2022) represent that when younger firms push out nonproductive older firms by increasing competition in the market. However, despite this in 2022 they also found that due to immigration restrictions that “older less productive firms are not being pushed out”. This further polarizes the market, also lowering skill premium. Mandelman et al. (2024) research the recent slowdowns in immigration and labor shortages that coincide with a declining skill premium. Their findings show that many countries in Latin America have recently had declining fertility rates, and recent policy changes in immigration have caused a shift in the distribution of skilled labor. This shift caused low-skilled immigrant labor to become less desired, increasing the amount of high-

skill labor coming into the U.S., whereas before, most workers from foreign lands worked on the lower half of the skill premium.

Furman and Seamans (2019) mention that while, on the one hand, there is insane productivity growth in the economy fixing the current problem of slowed productivity, there is the other side of the coin where AI can worsen the problem by causing a dip in the labor force participation rate. To combat this possible consequence, they propose policy solutions that include AI-specific commission, data portability, and a universal basic income (UBI). Their findings show that there is an uncertain prediction when weighing the tradeoffs. However, ways to fix this would be measuring the speed at which AI affects the economy and how it affects medium-, short-, and long-term growth.

Wang and Wang (2022) farther go over how AI has impacted the primary and secondary industries. They represent how AI has in-fact had a positive effect on industrial productivity, however, also consider risks. These risks include the ability to manipulate AI and how it can be used to destroy critical infrastructure in business since it is required an outside source of information. Where their research relates to this paper is in their conclusion. They state that while AI is very beneficial to society and has played a crucial role in productivity increase, there needs to be limitations and humanitarian policies.

Fiona Morton's review of *"Power and Progress"* by Dan Acemoglu and Simon Johnson gives an insight on if and how technology harms workers. She criticizes the paper by saying that the small elite become richer when new technology is introduced while the lower-class workers are stuck in poverty and eventually faded out of the workforce by said technology from frictions caused. To solve these frictions Morton introduces some policy ideas she thinks will work. These policies include ideas that would include heterogeneous groups of experts in policymaking to allow

robustness, as well as allow for some leeway of expertise in other fields. Another policy idea they go on to include is the idea of breaking up big tech, this would allow for a diversified market and for companies such as Meta to have more competition and improve due to said competitive market. Overall, Morton's review of Power and Progress aims to diversify the market and improve competition as well as sooth frictions when it comes to the integration of technology.

Data

The sources of the data are the Saint Louis Federal Reserve (FRED), the Census Bureau (CB), the Migration Policy Institute (MPI), and the Bureau of Labor Statistics (BLS). Some of the data has information from several years, such as the statistics from the BLS, while MPI is relevant to the study since it represents the previous skill distribution around the whole of the US for all people. This data allows us to represent the goal of this paper, which is to understand the relationship between AI and automation and foreign-born laborers.

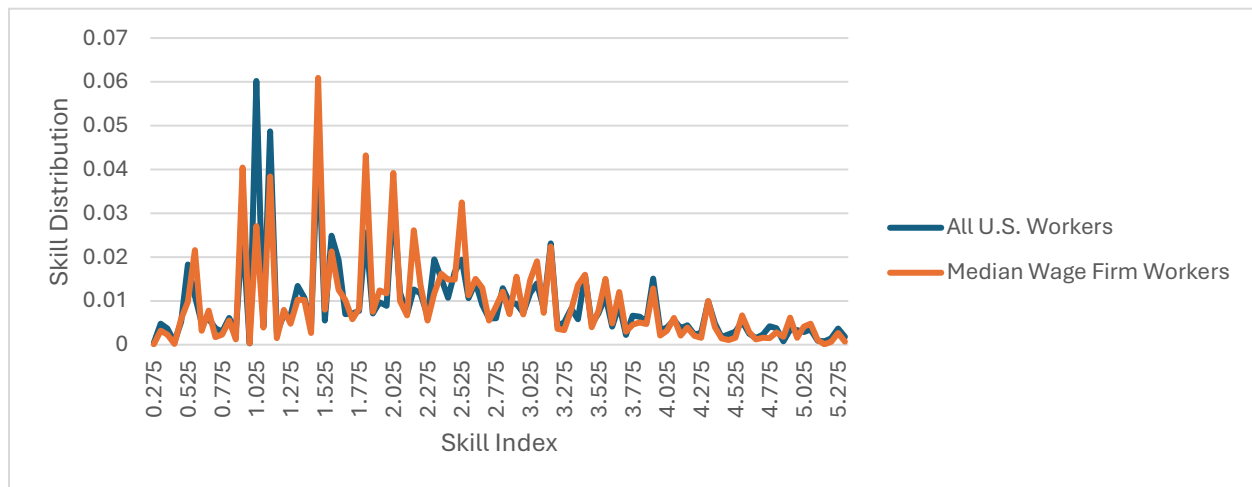
Data taken from the FRED is the foreign labor participation rate from January 2007 to May 2024. From MPI, we use the number of total immigrants in the U.S. measured between 1850 and 2022. Utilizing data from the Business Trends and Outlook Survey from the U.S., CB, we obtain information on the industries that use AI the most. The data is from October 23 to November 5, 2023.

Results

Figure 2 shows the skill distribution of all U.S. workers and the median wage firm workers using BLS data, which measures skill distribution from 1999. These results inform us that most workers are in the lower echelons of the labor force. This graph is for all workers in the

U.S. which also represents a possible risk that this technology isn't only damaging the immigrant workforce but the citizen workforce as well. However, when analyzing these job markets a high amount of immigrant or low-income labor is seen.

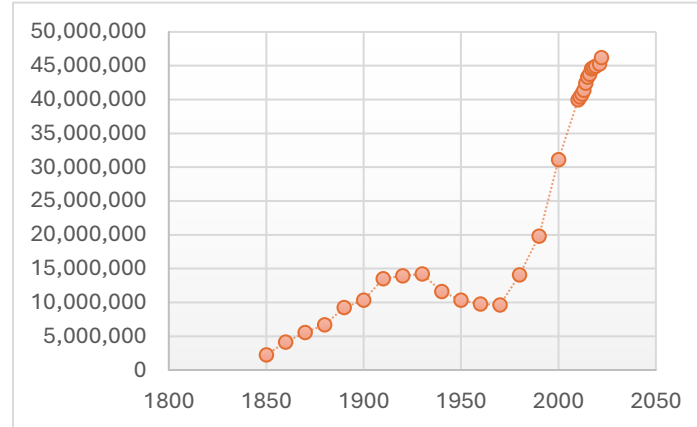
Figure 2: Skill Distribution 1999



Source: Cindy Michelle Cunningham and Robert D. Mohr, "Wage and job-skill distributions in the National Compensation Survey," *Monthly Labor Review*, U.S. Bureau of Labor Statistics, February 2017, <https://doi.org/10.21916/mlr.2017.3>

Additionally, according to Figure 3 immigration rates have only continued to rise exponentially. This fast-growing nation and its shrinking of the skill premium raises concerns mostly because of how these laborers are expected to contribute to our economy. If they are slowly fading out because of either automation or AI there needs to be certain policies put in place or new labor requirements.

Figure 3: Number of Immigrants in the Total US Population 1850 - 2022

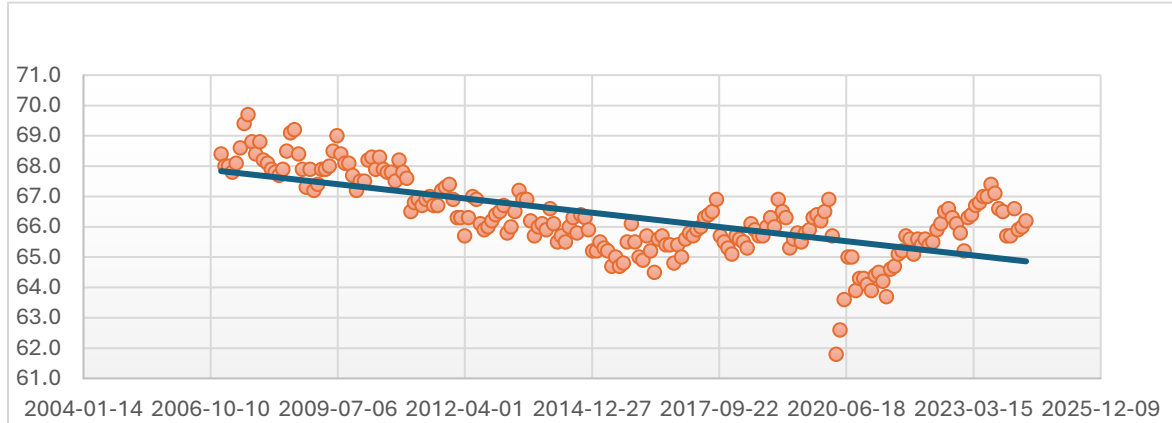


Source: "U.S. Immigrant Population and Share over Time, 1850-Present." *Migrationpolicy.Org*, 16 Oct. 2023,

www.migrationpolicy.org/programs/data-hub/charts/immigrant-population-over-time

The participation rate of the foreign-born labor force has been decreasing since the beginning of the 21st century. This can be due to many reasons, but immigration has not fallen. It's only grown, so why is this? Technological innovation has been seen over the years, especially regarding automation efficiency. As seen in Figure 4, as time progresses more and more into this technological age, the foreign-born labor participation rate decreases more and more. This can be attributed to things such as ChatGPT, assembly line robots, or just basic tools. The areas where AI and automation take the forefront are business and information, especially STEM work. Figure 4 represents this perfectly, showing and representing AI usage in different sectors and its expected growth leading into the next 6 months from when the study by the census was conducted in late November 2023.

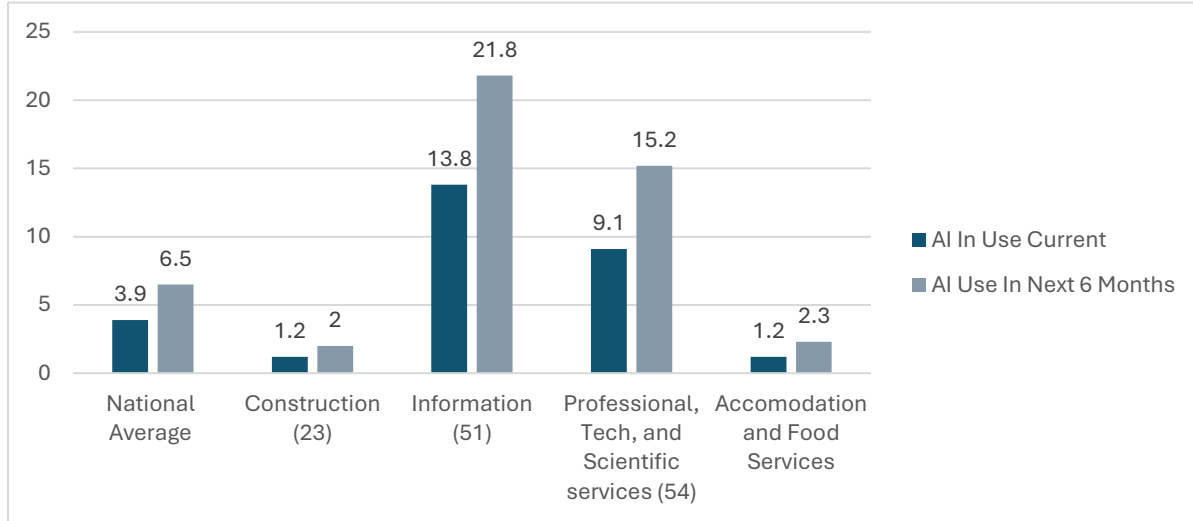
Figure 4: Foreign-Born Labor Force Participation Rate Over Time



Source: "Labor Force Participation Rate - Foreign Born." *FRED*, 7 June 2024, <https://fred.stlouisfed.org/series/LNU01373395>

Finally, Figure 5 depicts AI's (not automation's) rise in different sectors in the U.S. The two most changing of these two being information, tech, and other STEM fields. This is alarming because it represents a replacement for high skilled jobs too which an increase in higher skilled immigration has been seen as stated in the literature review, Mandelman and Zlate (2020). This sharp rise in these fields shows that there will be a farther compression of the labor distribution, most likely being towards middle skilled labor or labor that does not require as much education. There are issues that arise here that include the loss of complete job markets threatening computer science and data science industries the most so far.

Figure 5: AI's Rise in Use in Different Job Sectors



Source: Breaux, Cory and Dinlersoz, Emin. "Only 3.8% of Businesses Use AI to Produce Goods and Services, Highest Use in Information Sector." *Census.Gov*, 21 Dec. 2023, www.census.gov/library/stories/2023/11/businesses-use-ai.html

From the studies we analyzed and by connecting the trends seen in the data, we can state that there is a relationship between the new technological age, foreign labor, and the shift in skill premium. The methodology allows for this because these trends are easily recognizable in the graphical format because despite the immigration rate only increasing in the U.S., represented by the center for immigration studies in "*Foreign-Born Population Grew by 5.1 Million in the Last Two Years*" but yet despite this massive increase trends show their labor force participation rate is still declining. Whether this be today's technology or other policy restrictions, some conclusions can still be drawn from this data.

Conclusions

We conclude with some closing policy recommendations. From figure 4 and figure 5 there is a noticeable decline in immigrant workforce participation and an incline in the amount of AI used in certain job sectors. There are ways to combat this through policy either by imposing policy

restrictions on the amount of AI and automation being used especially with the recent developments at Tesla and Amazon, or by implementing less strict visa laws when it comes to immigrant businesses. These two policies could possibly help lessen the blow to the skill premium but especially when it comes to immigrant business create more competition forcing large corporations to innovate and not be as lazy.

Our recommendation for future research is to study the impact of immigrant business on competition not only innovation as well as what companies already have restrictions on their automated labor and how it impacts their growth. We believe that an empirical analysis done using the Difference in Differences (DID) approach could allow researchers to obtain causal estimates. The main challenge is to find a control group (county, industry, region, country) that has not been affected by automation or AI. The empirical findings could shed light on what the main mechanisms are to soon model them in a general equilibrium framework.

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